

Proposal for a dietary "phytochemical index".

There is ample reason to believe that diets rich in phytochemicals provide protection from vascular diseases and many cancers; direct antioxidant activity as well as modulation of enzyme expression or hormone activity contribute to this effect. Phytochemicals derived from diverse foods presumably can interact additively and (possibly) synergistically; thus, the total dietary load of phytochemicals may have important implications for health. As a means of very roughly quantifying this load, a "phytochemical index" (PI) is proposed, defined as the percent of dietary calories derived from foods rich in phytochemicals. Calories derived from fruits, vegetables (excluding potatoes), legumes, whole grains, nuts, seeds, fruit/vegetable juices, soy products, wine, beer, and cider - and foods compounded therefrom - would be counted in this index. Partial credit could be given for antioxidant-rich extra virgin olive oil. Other added oils, refined sugars, refined grains, potato products, hard liquors, and animal products - regrettably, the chief sources of calories in typical Western diets - would be excluded. Although the PI would provide only a very rough approximation of the quantity or quality of phytochemical nutrition, it nonetheless could aid epidemiologists in exploring the health consequences of diets high in phytochemical-rich plant foods, and could also help clinical nutritionists in their efforts to improve the phytochemical nutrition of their clients.